Module1: Introduction to Embedded System

Introduction to Embedded System: What is an Embedded Systems? Embedded systems Vs General computing systems, History of Embedded Systems, Classification of Embedded systems, Major Application Areas of Embedded Systems. Purpose of Embedded Systems, The Typical Embedded System, Microprocessor Vs Microcontroller, Differences between RISC and CISC, Harvard V/s Von- Neumann Processor/Controller Architecture, Big-endian V/s Little-endian processors, Memory (ROM and RAM types), Sensors & Actuators, The I/O Subsystem – I/O Devices, Light Emitting Diode (LED), 7-Segment LED Display, Optocoupler, Relay, Piezo buzzer, Push button switch, Communication Interfaces,On-board Communication Interface, External Communication Interface, Embedded Firmware, Other System Components
(Text 1: All the Topics from Ch-1 and Ch-2.)

Embedded system – Definition

An embedded system is an electronic/electro-mechanical system designed to perform a specific function and is a combination of both hardware and firmware (software).

Embedded systems are becoming an inevitable part of any product or equipment in all fields including household appliances, telecommunications, medical equipment, industrial control, consumer products, etc.

EMBEDDED SYSTEM & GENERAL PURPOSE COMPUTING SYSTEM

The Embedded System and the General purpose computer are at two extremes. The embedded system is designed to perform a specific task whereas as per definition the general purpose computer is meant for general use. It can be used for developing & playing games, watching movies, creating software, work on documents or spreadsheets etc. Following are certain specific points that differentiates between embedded systems and general purpose computers:

Criteria	General Purpose Computer	Embedded system
Contents	It is combination of generic hardware and a general purpose OS for executing a variety of applications.	It is combination of special purpose hardware and embedded OS for executing specific set of applications
Operating System	It contains general purpose operating system	It may or may not contain operating system.
Alterations	Applications are alterable by the user.	Applications are non-alterable by the user.
Key factor	Performance" is key factor.	Application specific requirements are key factors.
Power Consumption	More	Less
Response Time	Not Critical	Critical for some applications

CLASSIFICATION OF EMBEDDED SYSTEMS

The classification of embedded system is based on following criteria's:

- Based on generation
- Based on complexity & performance
- Based on deterministic behavior
- Based on triggering

Based on generation

1. First generation (1G):

- Built around 8bit microprocessor & microcontroller.
- Simple in hardware circuit & firmware developed.
- Examples: Digital telephone keypads, Stepper motor control unit

2. Second generation (2G):

- Embedded system built around 16bit Mp & 4,8,&16bit Mc.
- Complex & powerful Instruction set, Some ES contain embedded OS
- Examples SCADA systems, Data Acquisition System

3. Third generation (3G):

- Embeded system built around 32bit Mp & 16bit Mc.(Pentium, motorola 68000)
- Concepts like Digital Signal Processors (DSPs), Application Specific Integrated Circuits(ASICs) with instruction pipelining evolved.
- More Complex & powerful Instruction set
- To meet high performance requirement
- Examples: Robotics, Media, etc.

4. Fourth generation:

- Built around 64-bit µp & 32-bit µc.
- Concept of System on Chips (SoC), Multicore Processors evolved.
- To bring very high performance real time embedded market
- Makes use of very high performance real time embedded OS
- Examples: Smart phones, Mobile internet devices

Based on complexity & performance:

1. Small-scale:

- Simple in application need
- Performance not time-critical.
- Built around low performance& low cost 8 or 16 bit µp/µc.

Example: an electronic toy

2. Medium-scale:

- Slightly complex in hardware & firmware requirement.
- Built around medium performance & low cost 16 or 32 bit µp/µc.
- Usually contain operating system.
- Examples: Industrial machines.

3. Large-scale:

- Highly complex hardware & firmware.
- Built around 32 or 64 bit RISC µp/µc or PLDs or Multicore Processors.
- Response is time-critical.
- Examples: Mission critical applications.

<u>Based on deterministic behavior:</u>

- This classification is applicable for "Real Time" systems.
- The task execution behavior for an embedded system may be deterministic ornondeterministic.
- Based on execution behavior Real Time embedded systems are divided into Hard and Soft.

<u>Based on triggering</u>

- Embedded systems which are "Reactive" in nature canbe based on triggering.
- Reactive systems can be:
    - Event triggered
    - Time triggered

<u>APPLICATION OF EMBEDDED SYSTEM</u>

The application areas and the products in the embedded domain are countless.

1. Consumer Electronics: Camcorders, Cameras.
2. Household appliances: Washing machine, Refrigerator.
3. Automotive industry: Anti-lock breaking system(ABS), engine control.
4. Home automation & security systems: Air conditioners, sprinklers, fire alarms.
5. Telecom: Cellular phones, telephone switches.
6. Computer peripherals: Printers, scanners.
7. Computer networking systems: Network routers and switches.
8. Healthcare: EEG, ECG machines.
9. Banking & Retail: Automatic teller machines, point of sales.
10. Card Readers: Barcode, smart card readers.

PURPOSE OF EMBEDDED SYSTEM

1. **<u>Data Collection/Storage/Representation</u>**

- Embedded system designed for the purpose of data collection performs acquisition of data from the external world.
- Data collection is usually done for storage, analysis, manipulation and transmission. - Data can be analog or digital.
- Embedded systems with analog data capturing techniques collect data directly in the form of analog signal whereas embedded systems with digital data collection mechanism converts the analog signal to the digital signal using analog to digital converters.
- If the data is digital it can be directly captured by digital embedded system.
- A digital camera is a typical example of an embedded System with data collection/storage/representation of data.
- Images are captured and the captured image may be stored within the memory of the camera. The captured image can also be presented to the user through a graphic LCD unit.

2. **Data communication**

- Embedded data communication systems are deployed inapplications from complex satellite communication to simple home networking systems.
- The transmission of data is achieved either by a wire-lin medium or by a wire-less medium. Data can either be transmitted by analog means or by digital means.
- Wireless modules-Bluetooth, Wi-Fi.
- Wire-line modules-USB, TCP/IP.
- Network hubs, routers, switches are examples of dedicated data transmission embedded systems.

3. **Data signal processing**

- Embedded systems with signal processing functionalities are employed in applications demanding signal processing like speech coding, audio video codec, transmission applications etc.
- A digital hearing aid is a typical example of an embedded system employing data processing. Digital hearing aid improves the hearing capacity of hearing impaired person.

4. **Monitoring**

- All embedded products coming under the medical domain are with monitoring functions. Electro cardiogram machine is intended to do the monitoring of the heartbeat of a patient but it cannot impose control over the heartbeat.
- Other examples with monitoring function are digital CRO, digital multi-meters, and logic analyzers.

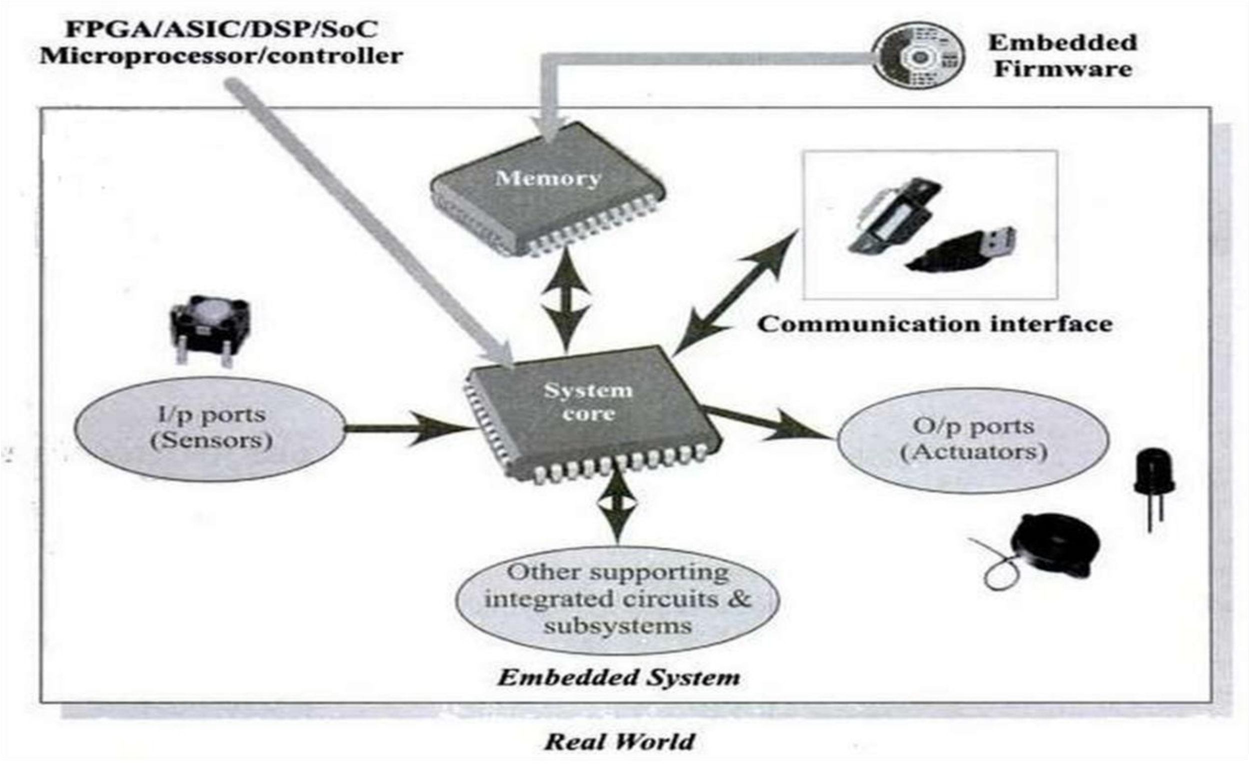
5. **Control**

- A system with control functionality contains both sensors and actuators Sensors are connected to the input port for capturing the changes in environmental variable and the actuators connected to the output port are controlled according to the changes in the input variable.
- Air conditioner system used to control the room temperature to a specified limit is a typical example for CONTROL purpose.

6. **Application specific user interface**

- Buttons, switches, keypad, lights, bells, display units etc are application specific user interfaces.
- Mobile phone is an example of application specific user interface.
- In mobile phone the user interface is provided through the keypad, system speaker, vibration alert etc.

ELEMENTS OF AN EMBEDDED SYSTEM



- A typical embedded system contains a single chip called system core, which acts as the master brain of the system.

- The system core of an embedded system can be a microprocessor or a microcontroller or a Field Programmable Gate Array (FPGA) device or a Digital Signal Processor (DSP) or an Application Specific Integrated Circuit (ASIC)/ Application Specific Standard Product (ASSP).
- Embedded hardware/software systems are basically designed to regulate a physical variable or to manipulate the state of some devices by sending some control signals to the actuators or devices connected to the o/p ports of the system, in response to the input signals provided by the end users or Sensors which are connected to the input ports. Hence an embedded sytem can be viewed as a reactive system.
- The control is achieved by processing the information coming from the sensors and user interfaces, and controlling some actuators that regulate the physical variable.
- Key boards, push button switches, etc. are examples for common user interface input devices.
- LEDs, liquid crystal displays, piezoelectric buzzers, etc. are examples for common user interface output devices for a typical embedded system.
- It should be noted that it is not necessary that all embedded systems should incorporate these I/O user interfaces. It solely depends on the type of the application for which the embedded system is designed.
- Some Embedded systems automatically sense the variations in the input parameters in accordance with the changes in the real world, through the sensors which are connected to the input port of the system. The processor performs some pre-defined operations on the sensed information with the help of the firmware embedded in the system and sends some actuating signals to the actuator connected to the output port of the embedded system.
- The Memory of the system is responsible for holding the control algorithm and other important configuration details. Most of embedded systems, uses ROM for storing the algorithm or configuration data and it is not available for the end user for modifications.
- The most common types of memories used in embedded systems for control algorithm storage are OTP, PROM, UVEPROM, EEPROM and FLASH. Depending on the control application, the memory size may vary from a few bytes to megabytes.
- Sometimes the system requires RAM for performing arithmetic operations or control algorithm execution. Various types of RAM like SRAM, DRAM and NVRAM are used for this purpose. The size of the RAM also varies from a few bytes to kilobytes or megabytes depending on the application.

  Different instruction set and system architecture are available for the design of a microprocessor. Harvard and Von-Neumann are the two common system architectures for processor design.

Von neuman architecture	Harvard architecture
Single address and data bus is used to connect program memory(ROM)& data memory(RAM)	Separate buses are used for connecting program memory(ROM)& data memory(RAM)
Accessing from data memory and program memory can not be done in parallel	Accessing from data memory and program memory can be done in parallel
Speed of execution is slow	Speed of execution is fast

 Reduced Instruction Set Computing (RISC) and Complex Instruction Set Computing (CISC) are the two common Instruction Set Architectures (ISA) available for processor design.

RISC	CISC
☐ Reduced Instruction Set Computing	☐ Complex Instruction Set Computing
☐ It contains lesser number of instructions.	☐ It contains greater number of instructions.
☐ Instruction pipelining and increased execution speed.	☐ Instruction pipelining feature does not exist.
☐ Orthogonal instruction set(allows each instruction to operate on any register and use any addressing mode.	☐ Non-orthogonal set(all instructions are not allowed to operate on any register and use any addressing mode.
☐ Operations are performed on registers only, only memory operations are load and store.	☐ Operations are performed either on registers or memory depending on instruction.
☐ A larger number of registers are available.	☐ The number of general purpose registers are very limited.
☐ Programmer needs to write more code to execute a task since instructions are simpler ones.	☐ Instructions are like macros in C language. A programmer can achieve the desired functionality with a single instruction which in turn provides the effect of using more simpler single instruction in RISC.
☐ It is single, fixed length instruction.	☐ It is variable length instruction.

Memory:

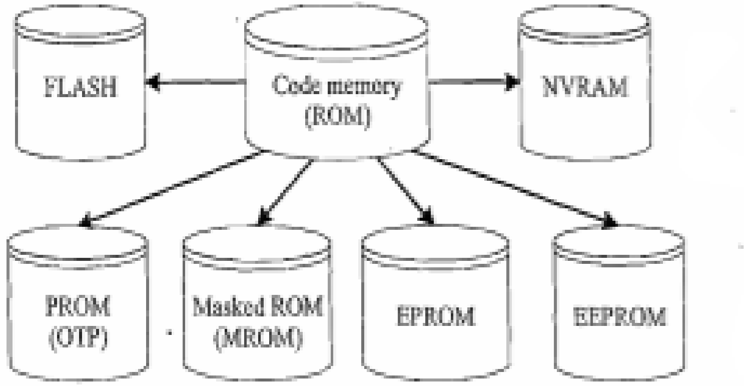
Memory is an important part of a processor/controller based embedded systems. If the memory is inbuilt in the processors/controllers, then this memory is referred as on-chip memory. If the processor/ Controller do not contain any memory inside and requires external memory to be connected with the controller/processor to store the control algorithm. It is called off-chip memory.

Types of Memory

Program Storage Memory (ROM)

The program memory or code storage memory of an embedded system stores the program instructions and it can be classified into different types depending on the fabrication, erasing and programming techniques.

Classification of Program Memory (ROM)



The code memory (ROM) retains its contents even after the power to it is turned off. It is generally known as non-volatile storage memory.

Masked ROM (MROM):

Masked ROM is a one-time programmable device. Masked ROM makes use of the hardwired technology for storing data. The device is factory programmed by masking and metallisation process at the time of production itself, according to the data provided by the end user. The primary advantage of this is low cost for high volume production.. Masked ROM is a good candidate for storing the embedded firmware for low cost embedded devices. The binary data corresponding to the proven design for which the firmware requirements are tested and frozen can be given to the MROM fabricator. The limitation with MROM based firmware storage is the inability to modify the device firmware against firmware upgrades. Since the MROM is permanent in bit storage, it is not possible to alter the bit information.

Programmable Read Only Memory (PROM) / (OTP):

Unlike Masked ROM Memory, One Time Programmable Memory (OTP) or PROM is not pre-programmed by the manufacturer. The end **user** is responsible for programming these devices. This memory has nichrome or polysilicon wires arranged in a matrix. These wires can be functionally viewed as fuses. It is programmed by a PROM programmer which selectively bums the fuses according to the bit pattern to be stored. Fuses which are not blown/burned represent logic “1” whereas fuses which are blown represents a logic “0”. The default state is logic “1”. OTP is widely used for commercial production of embedded systems whose proto-typed versions are proven and the code is finalised. It is a low cost solution for commercial production. Since OTPs cannot be reprogrammed, they are not useful and worth for development purpose.

Erasable Programmable Read Only Memory (EPROM):

Erasable Programmable Read Only Memory (EPROM) gives the flexibility to re-program the same chip several times as the code is subject to continuous changes during the development phase. EPROM stores the bit information by charging the floating gate of an FET. Bit information is stored by using an EPROM programmer, which applies high voltage to charge the floating gate. The stored information in EPROM can be erased by exposing it to ultra violet rays for a fixed duration through a quartz window. Even though the EPROM chip is flexible in terms of re-programmability, it needs to be taken out of the circuit board and put in a UV eraser device for 20 to 30 minutes. So it is a tedious and time-consuming process.

Electrically Erasable Programmable Read Only Memory (EEPROM):

The information contained in the EEPROM memory can be altered by using electrical signals at the register/Byte level. They can be erased and reprogrammed in-circuit. These chips include a chip erase mode and in this mode they can be erased in a few milliseconds. It provides greater flexibility for system design. The only limitation is their capacity is limited when compared with the standard ROM (A few kilobytes).

FLASH:

FLASH is the latest ROM technology and is the most popular ROM technology used in today's embedded designs. FLASH memory is a variation of EEPROM technology. It combines the re-programmability of EEPROM and the high capacity of standard ROMs. FLASH memory is organised as sectors (blocks) or pages. FLASH memory stores information in an array of floating gate MOS- FET transistors. The erasing of memory can be done at sector level or page level without affecting the other sectors or pages. Each sector/page should be erased before re-programming. The typical erasable capacity' of FLASH is 1000 cycles.

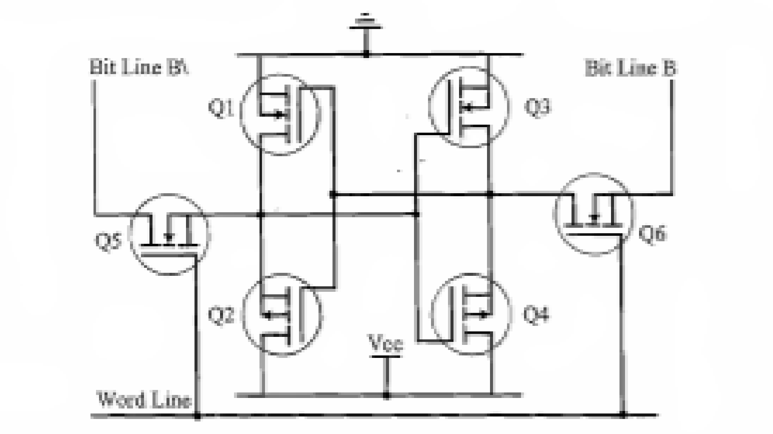
NVRAM:

Non-volatile RAM is a random access memory with battery backup. It contains static RAM based memory and a minute battery for providing supply to the memory in the absence of external power supply. The memory and battery are packed together in a single package. The life span of NVRAM is expected to be around 10 years. DS1644 from Maxim/Dallas is an example of 32KB NVRAM.

Read-Write Memory/Random Access Memory (RAM)

RAM is the data memory or working memory of the controller/processor. Controller/processor can read from it and write to it. RAM is volatile, meaning when the power is turned off, all the contents are destroyed. In a Random access memory, the desired memory location can be accessed directly without the need for traversing through the entire memory. RAM generally falls into three categories: Static RAM (SRAM), dynamic RAM (DRAM) and non-volatile RAM (NVRAM).

Static RAM (SRAM) Static RAM stores data in the form of voltage. They are made up of flip-flops. Static RAM is the fastest form of RAM available. In typical implementation, an SRAM cell (bit) is realised using six transistors (or 6 MOSFETs). Four of the transistors are used for building the latch (flip-flop) part of the memory cell and two for controlling the access. SRAM is fast in operation due to its resistive networking and switching capabilities. In its simplest representation an SRAM cell can be visualised as shown in Fig.



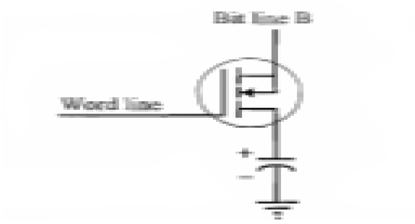
Access to the SRAM memory cell is controlled by Word Line, which controls the access transistors (MOSFETs) Q5 and Q6. The access transistors control the connection to bit lines B & B\.

In order to write a value to the memory cell, apply the desired value to the bit control lines (For writing 1, make B = 1 and B\ =0; For writing 0, make B = 0 and B\ =1) and assert the Word Line (Make Word line high). This operation latches the bit written in the flip-flop.

For reading the content of the memory cell, assert both B and B\ bit lines to 1 and set the Word line to 1. The major limitations of SRAM are low capacity and high cost.

Dynamic RAM (DRAM):

DynamicRAM stores data in the form of charge. They are made up of MOS transistor gates.The typical implementation of a DRAM cell is shown in fig. The MOSFET acts as the gate for the incoming and outgoing data whereas the capacitor acts as the bit storage unit.



The advantages of DRAM are its high density and low cost compared to SRAM. The disadvantage is that since the information is stored as charge it gets leaked off with time and the DRAM cell needs to be refreshed periodically. Special circuits called DRAM controllers are used for the refreshing operation. The refresh operation is done periodically in msecs interval.

Difference between SRAM and DRAM Cell

SRAM Cell	**DRAM Cell**
Made up of 6 CMOS transistors (MOSFET)	Made up of a MOSFET and a Capacitor
Low Capcity (Less Dense)	High Capcity (Highly Dense)
More Expensive	Less Expensive
Fast in operation. Typical access time is 10nS	Slow in operation due to refresh requirements. Typical access time is 60ns. Write operation is faster than read operation

SENSORS & ACTUATORS

An embedded system is in constant interaction with the Real world and the controlling / monitoring functions executed by the embedded system is achieved in accordance with the changes happening to the Real world.

The changes in system environment or variables are detected by the sensors connected to the input port of the embedded system. If the embedded system is designed for any controlling purpose, the system will produce some changes in the controlling variable to bring the controlled variable to the desired value.

It is achieved through an actuator connected to the output port of the embedded system. If the embedded system is designed for monitoring purpose only, then there is no need for including an actuator in the system.

For example, take the case of an ECG machine. It is designed to monitor the heart beat status of a patient and it cannot impose a control over the patient's heart beat and its order.

The sensors used here are the different electrode sets connected to the body of the patient. The variations are captured and presented to the user (may be a doctor) through a visual display or some printed chart.

Sensors

Sensor is a transducer that converts energy from one form to another by sensing the input for any measurement or control purpose.

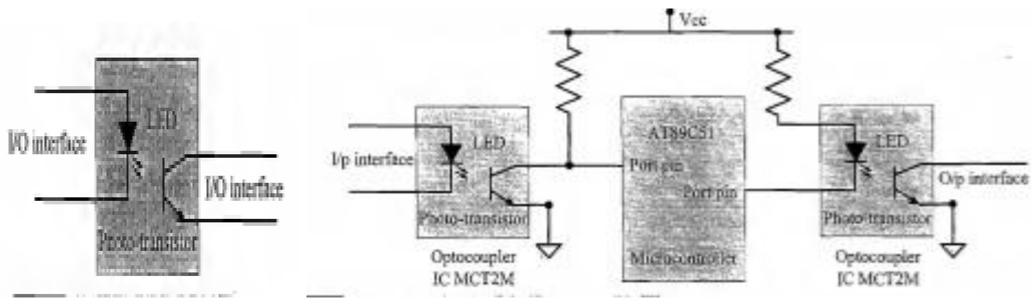
Ex. A Temperature sensor

Actuator

Actuator is used for output. It is a transducer that may be either mechanical or electrical which converts signals to corresponding physical actions.

Optocoupler:

Optocoupler is a solid state device to isolate two parts of a circuit. Optocoupler combines an LED and a photo-transistor in a single housing (package). In electronic circuits, an optocoupler is used for suppressing interference in data communication, circuit isolation, high voltage separation, simultaneous separation and signal intensification, etc. Optocouplers can be used in either input circuits or in output circuits as shown in fig.



COMMUNICATION INTERFACES

Communication interface is essential for communicating with various subsystems of the embedded system and with the external world.

For an embedded product, the communication interface can be viewed in two different perspectives; namely;

(i) Device/board level communication interface (Onboard Communication Interface)

(ii) Product level communication, interface (External Communication Inter face).

The communication channel which interconnects the various components within an embedded product is referred as device/board level communication interface (onboard communication interface).

Serial interfaces like I2C, SPI, UART, 1-Wire, etc and parallel bus interface, ptz examples of ‘Onboard Communication Interface’.

The communication channel responsible for data transfer between the embedded system and other devices or modules in distributed system is referred as Product level communication interface (External Communication Interface.).

The external communication interface can be either a wired media or a wireless media and it can be a serial or a parallel interface.

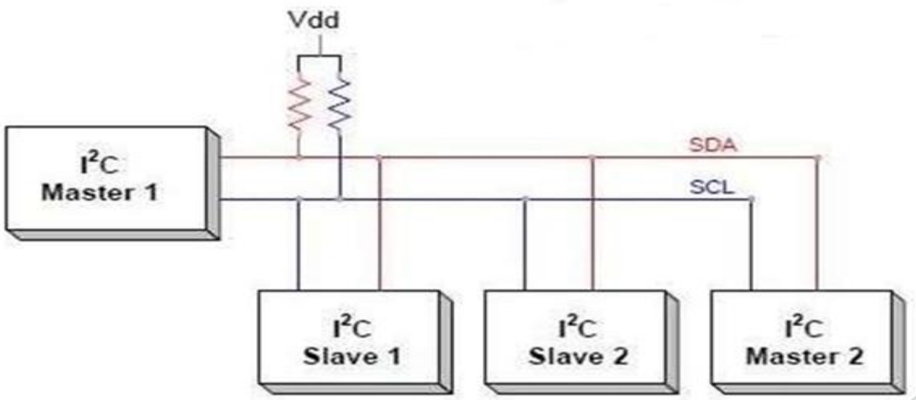
Infrared (IR), Bluetooth (BT), Wireless LAN (Wi-Fi), Radio Frequency waves (RF), GPRS, etc. are examples for wireless communication interface.

RS-232C/RS-422/RS-485, USB, Ethernet, IEEE 1394 port, Parallel port, etc. are examples for wired interfaces.

Onboard Communication Interfaces

Inter Integrated Circuit (I2C) Bus:

- A synchronous bi-directional half duplex (one-directional communication at a given point of time) two wire serial interface bus.
- The concept of I2C bus was developed by 'Philips semiconductors' in the early 1980s with an intention to provide an easy way of connection between a microprocessor /microcontroller system and the peripheral chips in television sets.
- The I2C bus comprise of two bus lines, namely; Serial Clock-SCL and Serial Data-SDA.
- SCL line is responsible for generating synchronisation clock pulses and SDA is responsible for transmitting the serial data across devices.
- I2C bus is a shared bus system to which many number of I2C devices can be connected in master slave configuration.
- Devices connected to the I2C bus can act as either 'Master' device or 'Slave' device. The 'Master' device is responsible for controlling the communication by initiating/terminating data transfer, sending data and generating necessary synchronisation clock pulses.
- 'Slave' devices wait for the commands from the master and respond upon receiving the commands. 'Master' and 'Slave' devices can act as either transmitter or receiver.
- Regardless whether a master is acting as transmitter or receiver, the synchronisation clock signal is generated by the 'Master' device only.
- I2C supports multi masters on the same bus. The following bus interface diagram shown in Fig. illustrates the connection of master and slave devices on the I2C bus.



The address of a I2C device is assigned by hardwiring the address lines of the device to the desired logic level.

The address to various I2C devices in an embedded device is assigned and hardwired at the time of designing the embedded hardware.

The sequence of operations for communicating with an I2C slave device is listed below:

1. The master device pulls the clock line (SCL) of the bus to ‘HIGH’
2. The master device pulls the data line (SDA) ‘LOW’, when the SCL line is at logic ‘HIGH’ (This is the ‘Start’ condition for data transfer)
3. The master device sends the address (7 bit or 10 bit wide) of the ‘slave’ device to which it wants to communicate, over the SDA line. Clock pulses are generated at the SCL line for synchronising the bit reception by the slave device. The MSB of the data is always transmitted first. The data in the bus is valid during the ‘HIGH’ period of the clock signal.
4. The master device sends the Read or Write bit (Bit value = 1 Read operation; Bit value = 0 Write operation) according to the requirement
5. The master device waits for the acknowledgement bit from the slave device whose address is sent on the bus along with the Read/Write operation command. Slave devices connected to the bus compares the address received with the address assigned to them
6. The slave device with the address requested by the master device responds by sending an acknowledge bit (Bit value = 1) over the SDA line
7. Upon receiving the acknowledge bit, the Master device sends the 8bit data to the slave device over SDA line, if the requested operation is ‘Write to device’. If the requested operation is ‘Read from device’, the slave device sends data to the master over the SDA line
8. The master device waits for the acknowledgement bit from the device upon byte transfer complete for a write operation and sends an acknowledge bit to the Slave device for a read operation
9. The master device terminates the transfer by pulling the SDA line ‘HIGH’ when the clock line SCL is at logic ‘HIGH’ (Indicating the ‘STOP’ condition).

I2C bus supports three different data rates of Standard mode (Data rate up to l00kbits/sec (100 kbps)), Fast mode (Data rate up to 400kbits/sec (400 kbps)) and High speed mode (Data rate up to 3.4Mbits/sec (3.4 Mbps)).

The first generation I2C devices were designed to support data rates only up to 100kbps. The new generation I2C devices are designed to operate at data rates up to 3.4Mbits/sec.

Serial Peripheral Interface (SPI) Bus:

The Serial Peripheral Interface Bus (SPI) is a synchronous bi-directional full duplex four-wire serial interface bus.

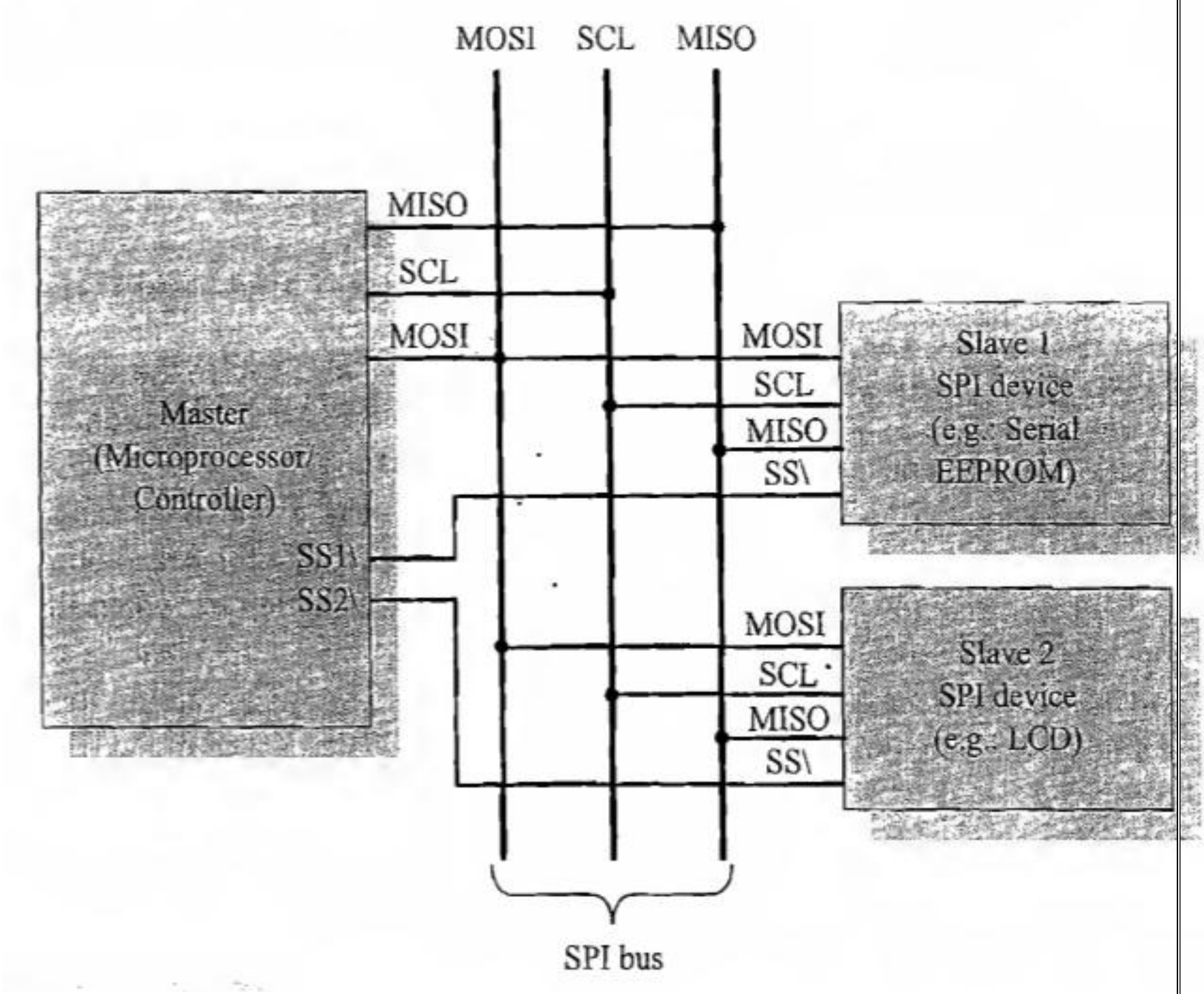
The concept of SPI was introduced by Motorola. SPI is a single master multi-slave system. It is possible to have a system where more than one SPI device can be master, provided the condition only one master device is active at any given point of time, is satisfied.

SPI requires four signal lines for communication namely

- **Master Out Slave In (MOSI):** Signal line carrying the data from master to slave device and is also known as Slave Input/Slave Data In (SI/SDI).

- Master In Slave Out (MISO): Signal line carrying the data from slave to master device and is also known as Slave Output (SO/SDO).
- Serial Clock (SCLK): Signal line carrying the clock signals.
- Slave Select (SS): Signal line for slave device select. It is an active low signal.
  - The bus interface diagram shown in Fig. illustrates the connection of master and slave devices on the SPI bus.
  - The master device is responsible for generating the clock signal.
  - It selects the required slave device by asserting the corresponding slave device's slave select signal 'LOW'.
  - The data out line (MISO) of all the slave devices when not selected floats at high impedance state.
  - The serial data transmission through SPI bus is fully configurable. SPI devices contain a certain set of registers for holding these configurations.
  - The serial peripheral control register holds the various configuration parameters like master/slave selection for the device, baudrate selection for communication, clock signal control, etc.
  - The status register holds the status of various conditions for transmission and reception. SPI works on the principle of 'Shift Register'.
  - The master and slave devices contain a special shift register for the data to transmit or receive. The size of the shift register is device dependent. Normally it is a multiple of 8.
  - During transmission from the master to slave, the data in the master's shift register is shifted out to the MOSI pin and it enters the shift register of the slave device through the MOSI pin of the slave device.

- At the same time the shifted out data bit from the slave device's shift register enters the shift register of the master device through MISO pin. In summary, the shift registers of 'master' and 'slave' devices form a circular buffer.



Infrared (IrDA)

- It is a serial, **half duplex**, line of sight based wireless technology for data communication between devices.
- It is in use from the olden days of communication. The remote control of TV, VCD player, etc works on Infrared data communication principle.
- Infrared communication technique uses infrared waves of the electromagnetic spectrum for transmitting the data. IrDA supports point-point and point-to-multipoint communication.
- The typical communication range for IrDA lies in the range 10 cm to 1 m. The range can be increased by increasing the transmitting power of the IR device.
- IR supports data rates ranging from 9600bits/second to 16Mbps. Depending on the speed of data transmission IR is classified into Serial IR (SIR), Medium IR (MIR), Fast IR (FIR), Very Fast IR (VFIR) and Ultra Fast IR (UFIR).

- SIR supports transmission rates ranging from 9600bps to 115.2kbps. MIR supports data rates of 0.576Mbps and 1.152Mbps. FIR supports data rates up to 4Mbps. VFIR is designed to support high data rates up to 16Mbps. The UFIR specs are under development and it is targeting a,data rate up to 100Mbps.
- IrDA communication involves a transmitter unit for transmitting the data over IR and a receiver for receiving the data.
- Infrared Light Emitting Diode (LED) is the IR source for transmitter and at the receiving end a photodiode acts as the receiver.
- Both transmitter and receiver unit will be present in each device(Transceiver) supporting IrDA communication for bidirectional data transfer.
- Certain devices like a TV remote control always require unidirectional communication and so they contain either the transmitter or receiver unit (The remote control unit contains the transmitter unit and TV contains the receiver unit).

Bluetooth

It is a low cost, low power, short range **wireless** technology for data and voice communication. Bluetooth supports point-to-point_ (device to device) and point-to-multipoint (device to multiple device broadcasting) wireless communication.

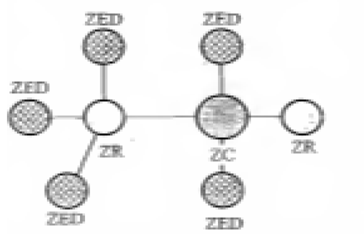
- A Bluetooth device can function as either master or slave. When a network is formed with one Bluetooth device as master and more than one device as slaves, it is called a **Piconet**. A **Piconet** supports a maximum of seven slave devices.
- Bluetooth is the favorite choice for short range data communication in handheld embedded devices. Bluetooth technology is very popular among cell phone users as they are the easiest communication channel for transferring music files, ring tones media files to neighboring Bluetooth enabled phones.
- It supports a data rate of **up to 1 Mbps** and a range of approximately **30 feet** for data communication.

Wi-Fi or Wireless Fidelity:

- It is the popular wireless communication technique for networked communication of devices. Wi-Fi is intended for network communication and it supports Internet Protocol (IP) based communication. It is essential to have device identities in a multipoint communication to address specific devices for data communication.
- **Wi-Fi based communications** require an intermediate agent called Wi-Fi router/Wireless access point to manage the communications. Wi-Fi supports data rates ranging from 1 Mbps to 150 Mbps and offers a range of 100 to 300 feet.

ZigBee:

- It is a low power, low cost, wireless network communication protocol based on the IEEE 802.15.4-2006 standard.
- ZigBee is targeted for low power, low data rate and secure applications for Wireless Personal Area Networking (WPAN).
- ZigBee operates worldwide at the unlicensed bands of Radio spectrum, mainly at 2.400 to 2.484 GHz, 902 to 928 MHz and 868.0 to 868.6 MHz.
- ZigBee supports an operating distance of up to 100 meters and a data rate of 20 to 250Kbps.
- ZigBee device categories are as follows:
- **ZigBee Coordinator (ZC)/Network Coordinator:** The ZigBee coordinator acts as the root of the ZigBee network. The ZC is responsible for initiating the ZigBee network and it has the capability **to store information about the network**.
- **ZigBee Router (ZR)/Full Function Device (FFD):** Responsible for passing information from device to another device or to another ZR.
- **ZigBee End Device (ZED)/Reduced Function Device (RFD):** End device containing ZigBee functionality for data communication



General Packet Radio Service(GPRS)

- General Packet Radio Service (GPRS) is a packet oriented mobile data service on the 2G and 3G cellular communication system's global system for mobile communications (GSM).
- GPRS was originally standardized by European Telecommunications Standards Institute (ETSI) GPRS usage is typically charged based on volume of data transferred, contrasting with circuit switched data, which is usually billed per minute of connection time. Some times billing time is broken down to every third of a minute. Usage above the bundle cap is charged per megabyte, speed limited, or disallowed.

Services offered:

- GPRS extends the GSM Packet circuit switched data capabilities and makes the following services possible:
- SMS messaging and broadcasting
- "Always on"internet access
- Multimedia messaging service (MMS)

- Push-to-talk over cellular(PoC)
- Instant messaging and presence-wireless village Internet applications for smart devices through wireless application protocol (WAP).
- Point-to-point (P2P) service: inter-networking with the Internet(IP).
- Point-to-multipoint (P2M) service:point-to-multipoint multicast and point-to- multipoint group calls.

Embedded Firmware

The control algorithm (Program instructions) and or the configuration settings that an embedded system developer dumps into the code memory of the embedded system The embedded firmware can be developed in various methods like

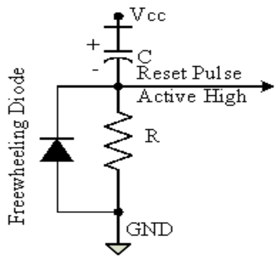
1. Write the program in high level languages like Embedded C/C++ using an Integrated Development Environment (The IDE will contain an editor, compiler, linker, debugger, simulator) etc. IDEs (Keil) are different for different family of processors/controllers.
2. Write the program in Assembly Language using the Instructions Supported by your application's target processor/controller

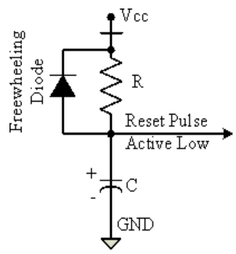
The process of converting the program written in either a high level language or processor/controller specific Assembly code to machine readable binary code is called 'HEX File Creation. The methods used for 'HEX File Creation ' is different depending on the programming techniques used. If the program is written in Embedded C/C++ using an IDE, the cross compiler included in the IDE converts it into corresponding processor/controller understandable 'HEXFile'.

Other System Components–

1. Reset Circuit

- The Reset circuit is essential to ensure that the device is not operating at a voltage level where the device is not guaranteed to operate, during system power ON.
- The Reset signal brings the internal registers and the different hardware systems of the processor/controller to a known state and starts the firmware execution from the reset vector (Normally from vector address 0x0000 for conventional processors/controllers.
- The reset vector can be relocated to an address for processors/controllers supporting boot loader.

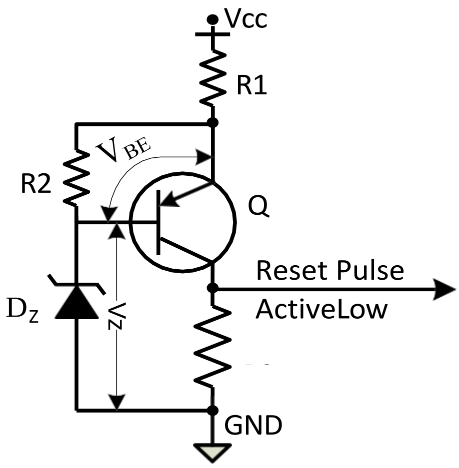




- The reset signal can be either active high (The processor undergoes reset when the reset pin of the processor is at logic high) or active low (The processor undergoes reset when the reset pin of the processor is at logic low).

2. Brown-out Protection Circuit

- Brown-out protection circuit prevents the processor/controller from unexpected program execution behavior when the supply voltage to the processor/controller falls below a specified voltage.
- The processor behavior may not be predictable if the supply voltage falls below the recommended operating voltage. It may lead to situations like data corruption.
- A brown-out protection circuit holds the processor/controller in reset state, when the operating voltage falls below the threshold, until it rises above the threshold voltage. Certain processors/controllers support built in brown-out protection circuit which monitors the supply voltage internally.



- If the processor/controller doesn't integrate a built-in brown-out protection circuit, the same can be implemented using external passive circuits or supervisor ICs

3. Oscillator Unit

- A microprocessor/microcontroller is a digital device made up of digital combinational and sequential circuits.
- The instruction execution of a microprocessor/controller occurs in sync with a clock signal. The oscillator unit of the embedded system is responsible for generating the precise clock for the processor.
- Quartz crystal Oscillators are example for clock pulse generating devices.

4. RealTimeClock(RTC)

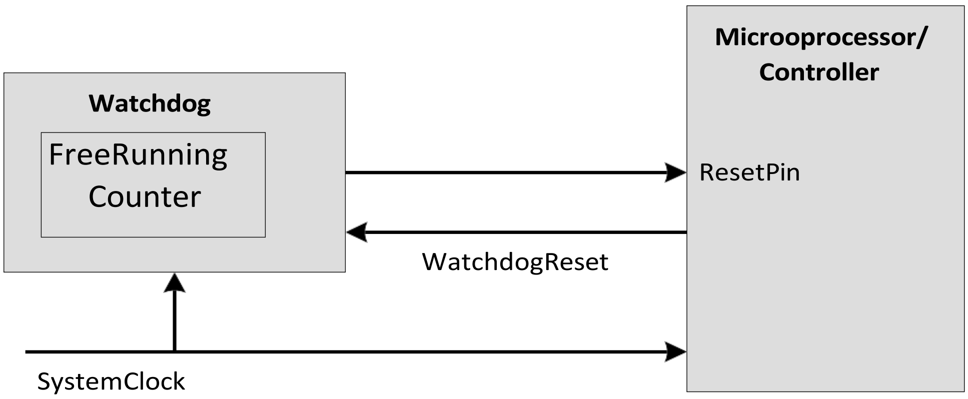
- It is a system component responsible for keeping track of time.
- RTC holds information like current time (In hour, minutes and seconds) in 12hour /24 hour

format, date, month, year, day of the week etc. and supplies timing reference to the system.

- RTC is intended to function even in the absence of power.
- RTCs are available in the form of Integrated Circuits from different semiconductor manufacturers like Maxim/Dallas, ST Microelectronics etc
- The RTC chip contains a microchip for holding the time and date related information and backup batterycellfor functioning intheabsenceofpower, inasingleICpackage.
- The RTC chip is interfaced to the processoror controller ofthe embedded system. For Operating System based embedded devices, a timing reference is essential for synchronizing the operations of the OS kernel.
- The RTC can interrupt the OS kernel by asserting the interrupt line of the processor/controller to which the RTC interrupt line is connected. The OS kernel identifies the interrupt in terms of the Interrupt Request (IRQ) number generated by an interrupt controller.
- One IRQ can be assigned to the RTC interrupt and the kernel can perform necessary operations like system date time updation, managing software timers etc when an RTC timer tick interrupt occurs.

5. Watch Dog Timer

- The watchdog timer is a timing device that resets the system after a predefined timeout.
- It is activated within the first few clock cycles after power-up.
- It helps in rescuing the system if a fault develops and program gets stuck.
- On restart, the system functions normally. The watchdog timer reset is a required feature in control applications.



- Depending on the internal implementation, the watchdog timer increments or decrements a free running counter with each clock pulse and generates a reset signal to reset the processor if the count reaches zero for a downcounting watchdog, or the highest count value for an up-counting watchdog.
- If the firmware execution doesn't complete due to malfunctioning, within the time required by

the watchdog to reach the maximum count, the counter will generate a reset pulse and this will reset the processor.

- If the firmware execution completes before the expiration of the watchdog timer the WDT can be stopped from action
- Most of the processors implement watchdog as a built-in component and provides status register to control the watchdog timer (like enabling and disabling watchdog functioning) and watchdog timer register for writing the count value.
- If the processor/controller doesn't contain a built-in watchdog timer, the same can be implemented using an external watchdog timer IC circuit.